Review of the Manuscript

An Evaluation and Comparison of GPU Hardware and Solver Libraries for Accelerating the OPM Flow Reservoir Simulator

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Summary:

The manuscript submitted to the *IEEE Transactions on Parallel and Distributed Systems (TPDS)* evaluates and compares the effectiveness of the application of multiple CPU and GPU architectures for the simulation of multiphase flow in porous media using the Open Porous Media Flow (OPM Flow) simulator. Additionally, a discussion about different implementations of linear solvers is also provided. The solver framework is tested on literature and proprietary reservoir models containing up to 1 million active cells. Finally, the results show significant but limited gains in terms of execution time, while some bottlenecks related to memory management in GPUs are identified.

Assessment:

The paper presents a relevant and timely comparative study of linear solver implementations in the OPM Flow simulator. The numerical experiments appear to be well-executed and meaningful, offering useful performance comparisons across different hardware configurations and solver implementations. The core contribution is valuable and could be of interest to the community.

However, the manuscript suffers from major deficiencies in structure, presentation, and language. It contains numerous grammatical errors and poorly constructed sentences that significantly hinder comprehension. The presentation of results requires substantial revision—tables include excessive data without clear labels, and figures often fail to convey the intended insights. These issues, along with a significant number of problems I identified throughout the manuscript, make it difficult to fully assess the quality of the results in the current version of the manuscript. A thorough revision of grammar, clarity, and overall presentation is necessary before resubmission.

Detailed remarks

Introduction

1. In general, this section would benefit of a more detailed literature review citing more papers in the field that have performed similar assessments of the performance of linear solvers as it is proposed in the manuscript.
2. (Page 1) *"Computational Flow Dynamics (CFD) simulation is becoming extremely important to speed up [...]"*
   CFD is typically referred to as *"computational* ***fluid*** *dynamics"*. Furthermore, to say that CFD is becoming relevant only now is inaccurate. CFD simulations has been around for decades.
3. (Page 1) *"Furthermore, to improve the realism of the simulation [...]"*
   What is meant by realism? Would that be accuracy?
4. (Page 1) *"This leads to bigger systems [...]"*
   Using "larger" instead of "bigger" would be more appropriate in this context. This issue is repeated multiple times throughout the paper.
5. (Page 1) *"The project is split into several modules, e.g., opm simulators [...]"*
   Giving an example of one of the modules is unnecessary.
6. *"(Page 1) [...] for example, they may even take 90% or more of the simulation time for certain models."*
   What is the reference for this claim?

7. (Page 1) *"Efficient iterative solvers available in DUNE library include GMRES and BiCGSTAB. [...] with examples including AMG, and the well-known ILU0 [...]"*
   Introduce the acronyms before using them. This kind mistake happens several times in the manuscript. Please revise where necessary. In addition, please add the appropriate references to the original works on DUNE, GMRES and BiCGSTAB.
8. (Page 1) *"[...] achieved by avoiding non-contributing computations."*
   What is a non-contributing computation? Please explain it in the text.
9. (Page 1) *"The Constrained Pressure Residual (CPR) [6] [...]"*
   Instead of citing Stone Ridge's website, it would be more appropriate to cite the original paper by Wallis, Kendall, and Little (DOI: 10.2118/13536-MS).
10. (Pages 1-2) *"It solves the mostly elliptic pressure system using an AMG preconditioner. To support large-scale [...]"*
    Everything starting at "To support large-scale" should be a separate paragraph. This kind of long sentence/paragraph appear many times in the text. Please revise when needed.
11. (Page 2) *"Develop a custom open-source bridge [...]"*
    What is a "bridge" in this context? Does this mean middleware?
12. (Page 2) *"Integrate both manual solvers [...]"*
    What is a *"manual solver"*?
13. (Page 2) *"Perform a thorough evaluation and comparison [...]"*
    What metric is going to be used for this? Time? Memory usage? Number of iterations? Please specifiy.

Background

14. (Page 2) *"Reservoir simulation [13], [14] [...]"*
    There are better references than OPM's wiki in [14].

15. (Page 2) *"Important rock properties are type, porosity, water saturation, and permeability."*
    The saturation of a phase is usually a simulation variable. Do the authors refer to the initial conditions of the medium?

16. (Page 2) *"The black oil model assumes three fluid phases [...] and three components [...]"*
    Explain the concept of a component as it might not be clear to all readers. In addition, briefly introduce the black-oil equations in this section.

17. (Page 2) *"For non-miscible flow [...]"*
    Use *"immiscible flow"* instead.

18. (Page 2) Equation (1) can be incorporated into the paragraph to make it easier to read. A suggestion would be: *"The oil pressure and water saturation are chosen, but the third variable may represent all three phases ($\boldsymbol{s}_g$), no gaseous phase ($\mathbf{r}_{go}$) or no oleic phase ($\mathbf{r}_{og}$)."*

19. (Page 3) "*Figure 1 shows the general structure of OPM Flow.*"
    Figure 1 looks more like a flowchart of the simulation algorithm implemented by OPM Flow. Perhaps a more accurate description would be *"the main steps in the OPM Flow simulation workflow"*. Please update the reference to the figure and its caption.

20. (Page 3) *"The reservoir is modeled by a grid [...]"*
    This sentence is a bit too generic. A more precise formulation would be "the computational domain of the reservoir is discretized via a mesh".

21. (Page 3) *"For Irregular Corner-Point grids, coordinates lines or pillars are given to indicate x and y coordinates. Then, the top and bottom surfaces are specified by the z-coordinates of the cell's corner points along the for adjacent pillars."*
    From the description, the *"irregular"* corner-point grid sounds like the same as a typical corner-point grid. What would make it irregular? Please explain in the paragraph or just refer to it as a corner-point grid.

22. (Page 3) *"The radial grids are currently modeled as 2D cylindrical grids because no flow in the theta direction is implemented."*
What is the theta direction? Please introduce the term before using.

23. (Page 3) *"Grids cannot be combined in OPM."*
Does this mean that one cannot use multiple types of grids or that two distinct grids cannot be merged into one? Please ellaborate.

24. (Page 3) *"[...] M and N are 4 and 3 for blackoil respectively."*
It is clear that these are the values for M and N, but what do these numbers mean? Is M the number of variables and N the number of phases? Please clarify in the text.

25. (Page 3) *"[...] with unitriangular matrix L [...]"*
The term *"unitriangular"* is not standard. To make it clearer, it is better to refer to L as a lower triangular matrix with ones on the diagonal.

26. (Page 3) *"[...] which is equivalent to solving (1) [...]"*
Equation (1) does not represent a system of equations. Either correct the reference or rewrite it to make clear what equations are being solved.

27. (Page 4) *"In OPM terminology, this is called an ILU0 application [...]"*
The phrasing here is a bit awkward since using the ILU-0 procedure is called an application everywhere.

28. (Page 4, Algorithm 1) The algorithm does not describe anything and could be omitted.

29. (Page 4, Algorithm 2) The algorithm could have been much more descriptive. First, the output is the incomplete LU factorization. Thus, saying that it is "matrix A, but contains L and U" makes it confusing. Next, the notation a_{ii} is not introduced before using. What does it refer to? Also the variable S is never initialized or introduced. Finally, as written, the first if statement is empty, i.e., nothing happens if the condition is met. Is it just bad indentation? Please revise.

30. (Page 4) *"GPUs are traditionally designed for display purposes."*
This is poorly written. A better phrasing would be *"GPUs were originally designed for graphical applications"*.

31. (Page 4) *"[...] if the processing algorithm has enough parallelism."*
What does it mean to *"have"* parallelism? Make it clearer.

32. (Page 4, Table 1) This table could be removed from the paper as it does not convey any relevant information for the reader.

Implementation

33. (Page 5) *"When the wells are separate, the linear operation (spmv) [...]"*
For completeness, please define what spmv means and what operation it performs.

34. (Page 5) *"[...] the elements of A are actually small, dense blocks of size NxN [...]"*
Define what N is here. If it is the same as in page 3, then just reinforce this here for clarity.

35. (Page 6) *"[...] if the partitioning is done in a smart way (using the transmissibilities)."*
Briefly explain how the transmissibility values could be used for the partitioning.

36. (Page 6, Algorithm 3) This algorithm is unnecessary. The information presented in it is simple enough to be described in the text.

37. (Page 6, algorithms 4 and 5) It might be easier to understand these algorithms if they were re-written as equations. Since these are simple enough and essentially illustrate an operation that is being applied to each well, the procedure can be described as an equation.

38. (Page 6) *"For D, the inverse D^{-1} is stored instead, since multiplying with the inverse is easier."*
Why is this the case? What properties D has that it is easier to apply its inverse? Please elaborate.

39. (Page 7, Figure 2) Move this figure closer to its reference. The first mention to it happens two pages before it is shown. In addition, use a more descriptive caption. What are the numbers in each side? Perhaps a smaller example would be easier to understand.

Experimental Results

40. (Page 7) *"[...] in which the grid size is increased [...]"*
Probably this means that the grid is refined. Please be clearer if this is the case, as increasing the size of the grid could also mean increasing the size of the elements, thus reducing the resolution, or increasing the overall size of the computational domain.
41. (Page 7) *"Finally, the last benchmark used is proprietary [...]"*
Even though the data from this benchmark is not publicly available, it would be nice to briefly describe the heterogeneities in the model. This would give an idea of how hard it is solve it besides just the grid resolution.
42. (Page 7, Figure 4) The notation used here is overly complicated and makes it very hard for the reader to understand the results. Simplify the steps here and use charts to visualize the time of simulation per stage.
43. (Page 8, Figure 3) This figure is never referenced in the text.
44. (Page 8) *"Furthermore, when a certain date is mentioned, the latest merge commit before that date/time is used to perform the run."*
This information is unnecessary and could be omitted.
45. (Page 8) *"Simula [37] is a Norse research institute. Its main activities [...]"*
This information is not important in this context.
46. (Page 8) *"[...] to compare how the new hardware from Simula performs against our previous work [...]"*
What previous work? Please add a reference here.

47. (Page 8) *"[...] we add an optimization technique for ILU0 preconditioner [...]"*
    What kind of optimization is being done here? Please elaborate.

48. (Page 9) *"[...] the experiments with an out-of-the-box configuration revealed a surprising 1.35x slowdown [...]"*
    Explain why this results is surprising.

49. (Page 9) *"However, an equivalent analysis can be performed for the other two use cases with approximately similar conclusions. This is left as an exercise for the reader because this can be easily inferred from Table VIII."*
    The second sentence ("This is left as an exercise [...]") is inadequate. You can remove it and keep just the first sentence ("However, an equivalent analysis [...]").

50. (Page 10) "*The first thing we notice is the contrast between the scale_bsrsv2 [...]*"
    What does the "scale_bsrsv2" do? Be more specific.

51. (Page 10) *"[...] the bandwidth is almost doubled from 118 GB/s to 196 GB/s [...]"*
    This sentence is misleading. The aforementioned increase is approximately equal to 66% or 1.66 times, not close to the double. Please rectify the text.

52. (Page 10) *"Please note that due to issues with running omniperf for NORNE and NORNE modified, we report omniperf profile numbers only for the bigmod use case."*
    What issues were identified? Please report the problems that were encountered.

53. (Page 11, Table X) This table is being referenced before Table IX. Swap the order of the labels or the position of the tables in the text.

54. (Page 11, Table IX) The labels used for the rows in this tables are very confusing. Please describe what each operation means, as the reader most likely does not know what each function does.

55. (Page 11) *"Therefore, one of the key conclusions to optimize is to reduce the cost for the memory transfers and copy (sic) to jacobi matrix."*
    The cost of the memory access itself cannot be changed as it is hardware-specific. You probably mean to reduce the number of accesses instead.

Appendix A

56. If no comparisons with other libraries and simulators are made, this appendix can be omitted. It does not contribute to the manuscript besides acknowledging other works, which could be done in the introduction.